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A Unified Platform to Evaluate STDP Learning Rule and Synapse Model using Pattern Recognition in a Spiking Neural Network

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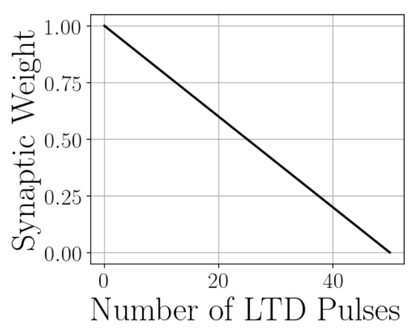
Overview

- We evaluate **four Spike Time Dependent Plasticity (STDP)** rules and **Three Synapse Models** on **MNIST Dataset.**
- Neuron Model: **Leaky Integrate and Fire** with **Adaptive Thresholding.**
- Synapse Models:
  - **Ideal Synapse**
  - **Linear Synapse**
  - **Non-Linear Synapse**
- STDP Models: **Ideal** vs. **Trigonometric** vs. **nGauss**
- **Python package** for simulation

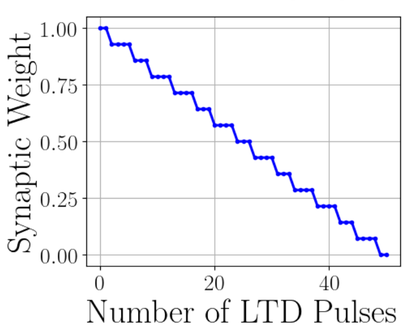
Synapse Models

Ideal Synapse



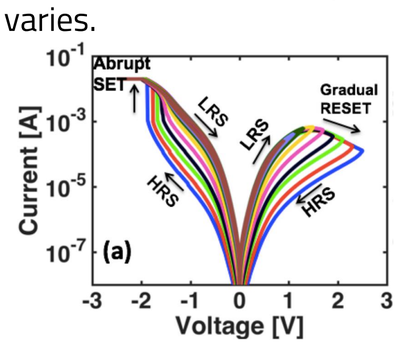
Weight can be anything between 0 and 1

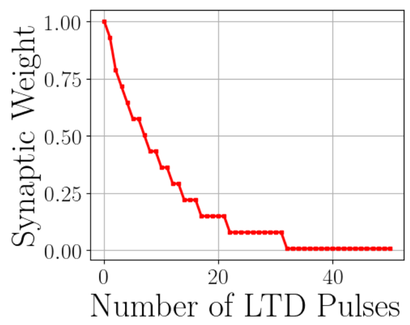
Linear Synapse



There are n states between 0 and 1

Non-Linear Synapse There are still n states between 0 and 1, but number of pulses needed for weight change varies.

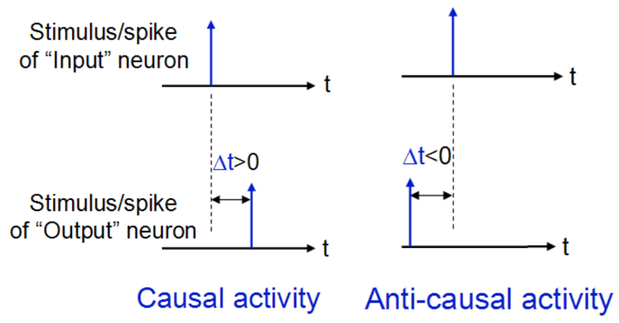




Derived from $Pr_{0.7}Ca_{0.3}MnO_3$ memristor data [3]

What are Spiking Neural Networks ?

- Fastest Super Computer: **Frontier (21.1 MWatt)**
  $$62.86 \times 10^9\ ops/Watt^{[1]}$$
- Human Brain: **Inside us (20 Watt)**
  $$\approx 50000 \times 10^9 ops/Watt^{[2]}$$
- Input is encoded as a spike train, two layer fully connected network.
- Weight represent the strength of connection between two neurons
- How is it different from ANNs ?
  - **No BACKPROPOGATION!**
  - Information encoded over time
- How does learning happen ?
  - **Spike Timing Dependent Plasticity (STDP)**



MNIST Encoding

Normalize the 28 x 28 image, and convert each pixel's value to its frequency. Fire that particular pixel at that frequency.

$$f = p \times (f_{\{max\}} - f_{\{min\}}) + f_{\{min\}}$$

$f_{\{max\}} = 70\ Hz, f_{\{min\}} = 5Hz$ with duration of each spike train as 100. Spike train for MNIST is 784 x 100 matrix.

M[i][j] = 1 implies i^{th} neuron fires at j^{th} timestep.

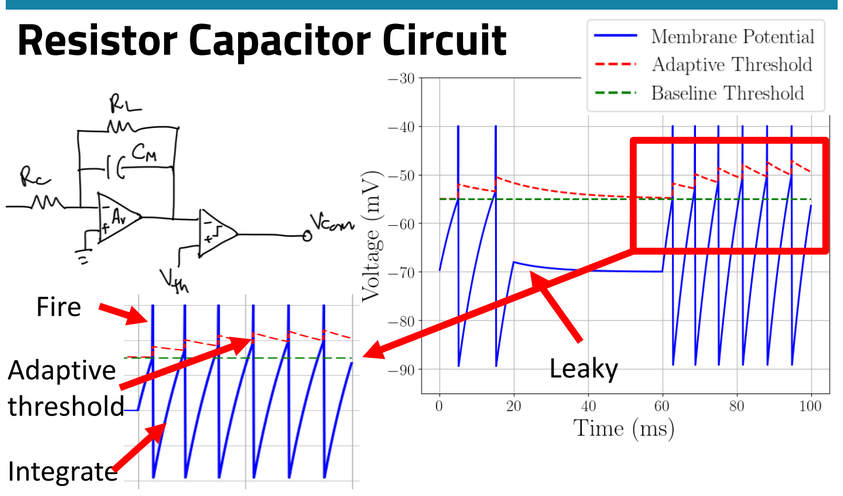
Corresponding CNN and Some Parameters

CNN with two convolutional layers followed by max pooling, a flattening step, and two fully connected layers.

Number of training images for both CNN and SNN: **100**
Number of testing images for both CNN and SNN: **1500**

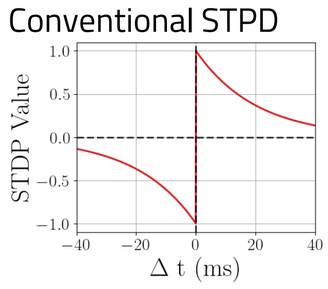
Leaky Integrate and Fire Neuron

Resistor Capacitor Circuit

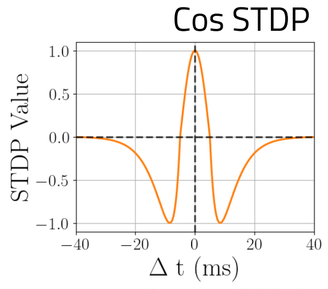


STDP Models

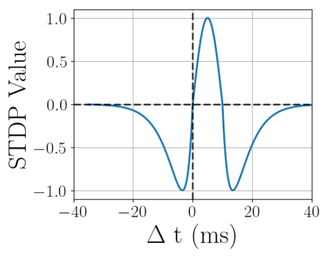
Conventional STPD



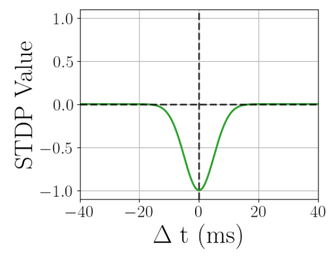
Cos STDP



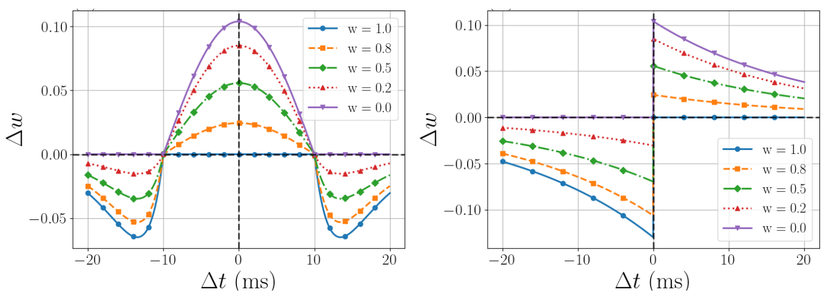
Sin STDP



nGauss STDP



Weight change for an ideal synapse using Cos STDP and Conventional STDP



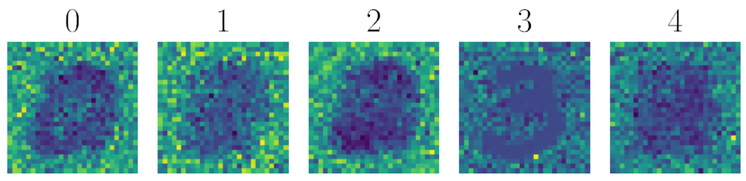
Algorithm and Python Package

```
for each neuron, get spike train
    pass spike train through network
        X = set of fired neurons in layer 2
        Y = set of contributing neurons for X
            in layer 1
        strengthen_synapses(X,Y)
        weaken_synapses(all_neurons - X , Y)
```

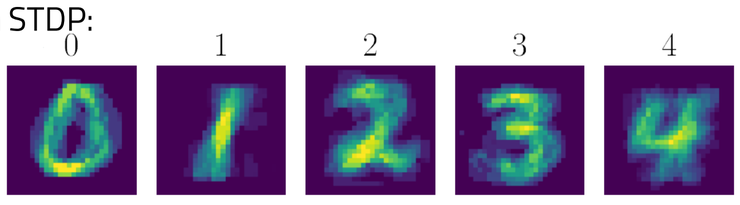
Package name: **nervos** (pip install nervos)
Includes code from scratch, with MNSIT and IRIS as examples.

Learned Weights for Cos and Sin STDP

Cos STDP:



Sin STDP:



Conclusion

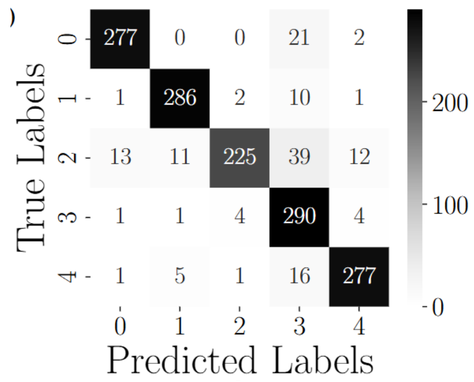
- Best rule for static pattern recognition: **Conventional STDP** because it captures the causal relationship between the neurons.
- Lowering the number of states in synapse leads to loss in accuracy.
- Other STDP might be useful in temporal encoding scheme, such as **Audio Recognition.**
- Requires less training data, architecture, and less epochs to convergence.

Results

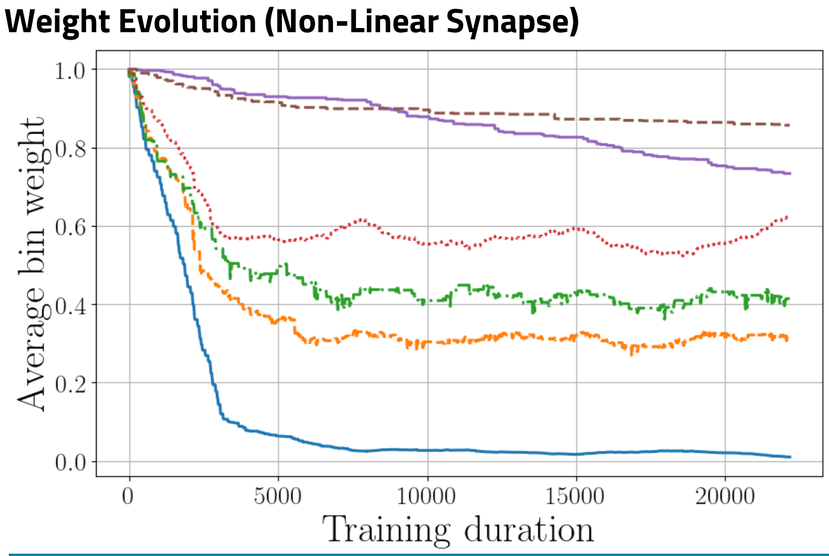
Changing STDP Rules

Number of Classes	Accuracy (%)		
	Conventional STDP	Cos STDP	Sin STDP
5	92.73	25.72	86.40
7	87.32	17.60	80.44
10	76.27	13.45	69.86

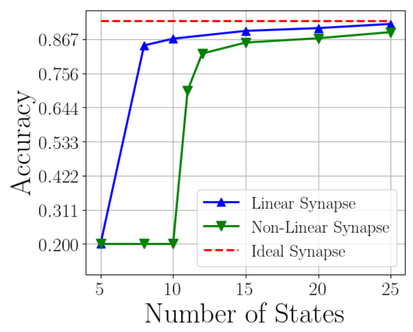
Confusion Matrix for Conventional STDP



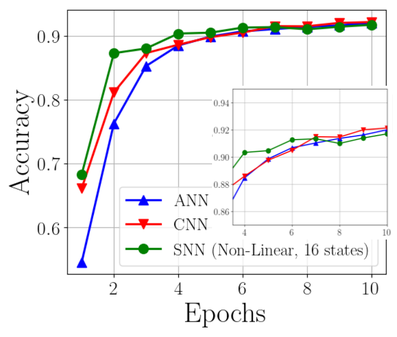
Weight Evolution (Non-Linear Synapse)



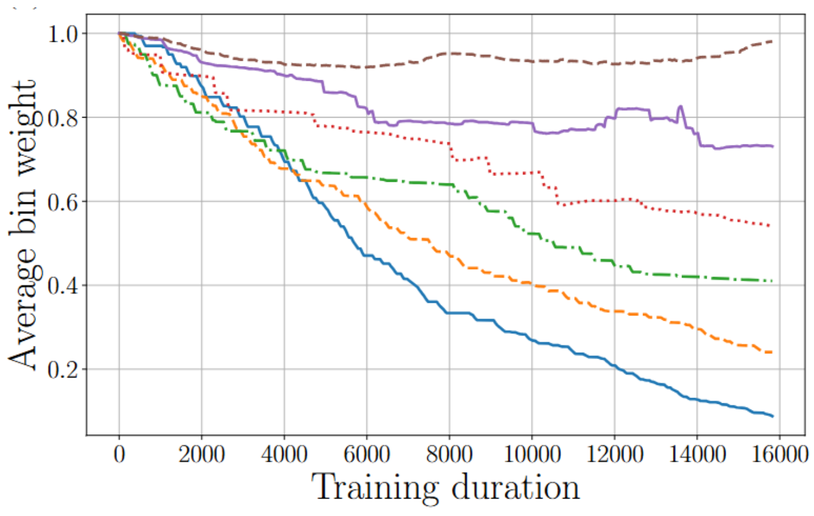
Changing Synapse Model



Comparison with CNN



Weight Evolution (Ideal Synapse)



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3. https://doi.org/10.1109/vlsi-tsa48913.2020.9203710

Acknowledgement

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